

MUHAMMEDH HAMDHAN

Address: 72, Katugasthota Road, Kandy

Phone: +94 772 994 015

Email: muhammedhhamdhan985@gmail.com

GitHub: <https://github.com/Hamdhan-10>

SUMMARY

Self-motivated and detail-oriented undergraduate in Computing with hands-on experience in developing **software applications, web applications, mobile/desktop apps, and IoT-based systems** using Python, Java, C, C#, Laravel, and Arduino. Skilled in coding, interface design, and testing across academic and personal projects. Passionate about solving real-world problems through innovative and efficient software solutions. Currently pursuing higher studies in Computing with a strong foundation in software engineering, and eager to contribute my technical knowledge and creative thinking to a forward-thinking IT or tech-driven organization.

CORE COMPETENCIES

- **Development Practices:** OOP, Testing, Wiring, Console Applications
 - **Technical Skills:** Python, Java, C, C#, MySQL, Laravel, JScript
 - **Tools & Technologies:** VS Code, Arduino IDE, GitHub
 - **Soft Skills:** Management, Creativity, Leadership, Critical Thinking, Negotiation
-

EDUCATION

- **Higher National Diploma (HND) in Computing** – NIBM (*Ongoing*)
 - **Diploma in Software Engineering** – NIBM (*Completed; awaiting transcript*)
 - **Certificate Course in Software Engineering** – NIBM (*Completed*)
 - **G.C.E. Ordinary Level** – AL-Hudha International School
-

EXTRA CURRICULAR ACTIVITIES

- **Senior Head Prefect** — AL-Hudha International School
 - **Football Team Captain** — AL-Hudha International School
-

PROJECT EXPERIENCE

1. Hand Gesture Recognition System

Developed a system that translates sign language gestures into text and speech using **Python, OpenCV, and MediaPipe**.

- Contributed in **organizing, coding, and designing** the project workflow.
- Coded the **hand detection module** that enabled data collection for training machine learning models.

- Designed the system to ensure real-time accuracy and usability.

2. Smart Farm (IoT-based)

Created an IoT-driven smart farm system using **Arduino and sensors** to monitor soil conditions and automate tasks.

- Designed the farm layout and user interface for system monitoring.
- Assisted in integrating sensors to improve farming efficiency and sustainability.

3. Inventory Management System

Developed a **Java-based inventory system** with backend database support for stock and transaction management.

- Designed and coded the **Graphical User Interface (GUI)** for user-friendly operation.
- Implemented backend logic and **MySQL Workbench database integration**.
- Ensured data accuracy through error-handling and validation techniques.

4. Intruder Detection System

Built a motion-based security solution using **Arduino and motion sensors** to detect unauthorized movements.

- Worked on **structural design and circuit implementation**.
- Integrated an alert mechanism (buzzer/light) to notify about intrusions.
- Conducted testing to minimize false detections.

5. Web Application with CRUD Operations

Designed and developed a **Laravel-based web application** with user authentication and CRUD functionalities.

- Built a responsive user interface with Blade templates.
- Integrated MySQL database for dynamic data storage and retrieval.
- Implemented role-based access and user-friendly navigation.

6. VPN Design (Networking Project)

Designed and simulated a **Virtual Private Network (VPN)** using **Cisco Packet Tracer**.

- Configured routers, firewalls, and tunneling protocols for secure communication.
- Tested secure data transfer between remote office networks.
- Focused on implementing practical networking and cybersecurity concepts.